

The Co-Area Mesh Method (CMM)

endogenous grid points on price planes, $h = 0$ enforced by construction

THE IDEA (user's proposal):

Standard solvers fix grid points (u_1, u_2, u_3) in the signal cube and treat prices $P[i, j, l]$ as unknowns. The co-area evidence integral

$A_v(p) = \int_{\{P=p\}} f_v f_v d\sigma$ then requires either:

- kernel smoothing $K_h(P - p) \rightarrow$ bandwidth $h > 0$, biased
- pointwise contour extraction $\rightarrow$ non-smooth in P , Newton fails

CMM inverts the representation: choose price levels $p_1 < \dots < p_M$ FIRST, and let the GRID POINTS be the unknowns. Each level surface $S_m = \{P = p_m\}$ is a triangle mesh whose vertex positions move. Grid points live exactly ON planes of prices -- endogenous coordinates.

The evidence integral becomes an arclength integral along the curve $S_m \cap \{u_k = U\}$: EXACT at $h = 0$. No kernel. No bandwidth. The "h $\rightarrow$ 0 limit" is not taken; it is enforced by the representation.

EQUATIONS: at every mesh vertex v of surface S_m :

$$\text{clear_CRR}(\mu_1(v), \mu_2(v), \mu_3(v); \gamma) = p_m$$

where $\mu_k(v)$ is the Bayes posterior from the slice integrals.

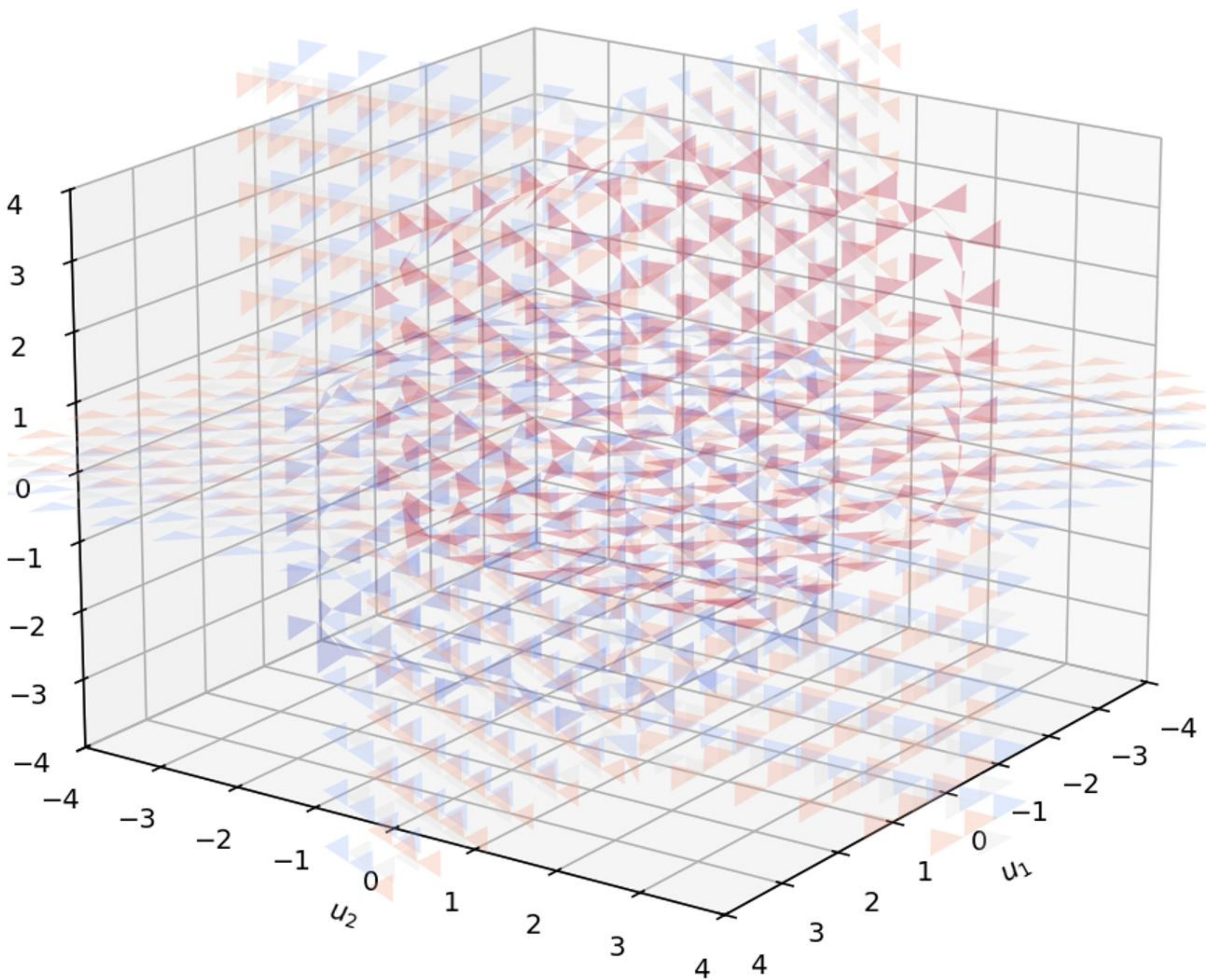
Unknowns: vertex positions (3 DOF each). Equations: 1 per vertex.

The 2-dimensional tangential slack per vertex is gauge freedom (sliding along the surface), absorbed by Levenberg damping.

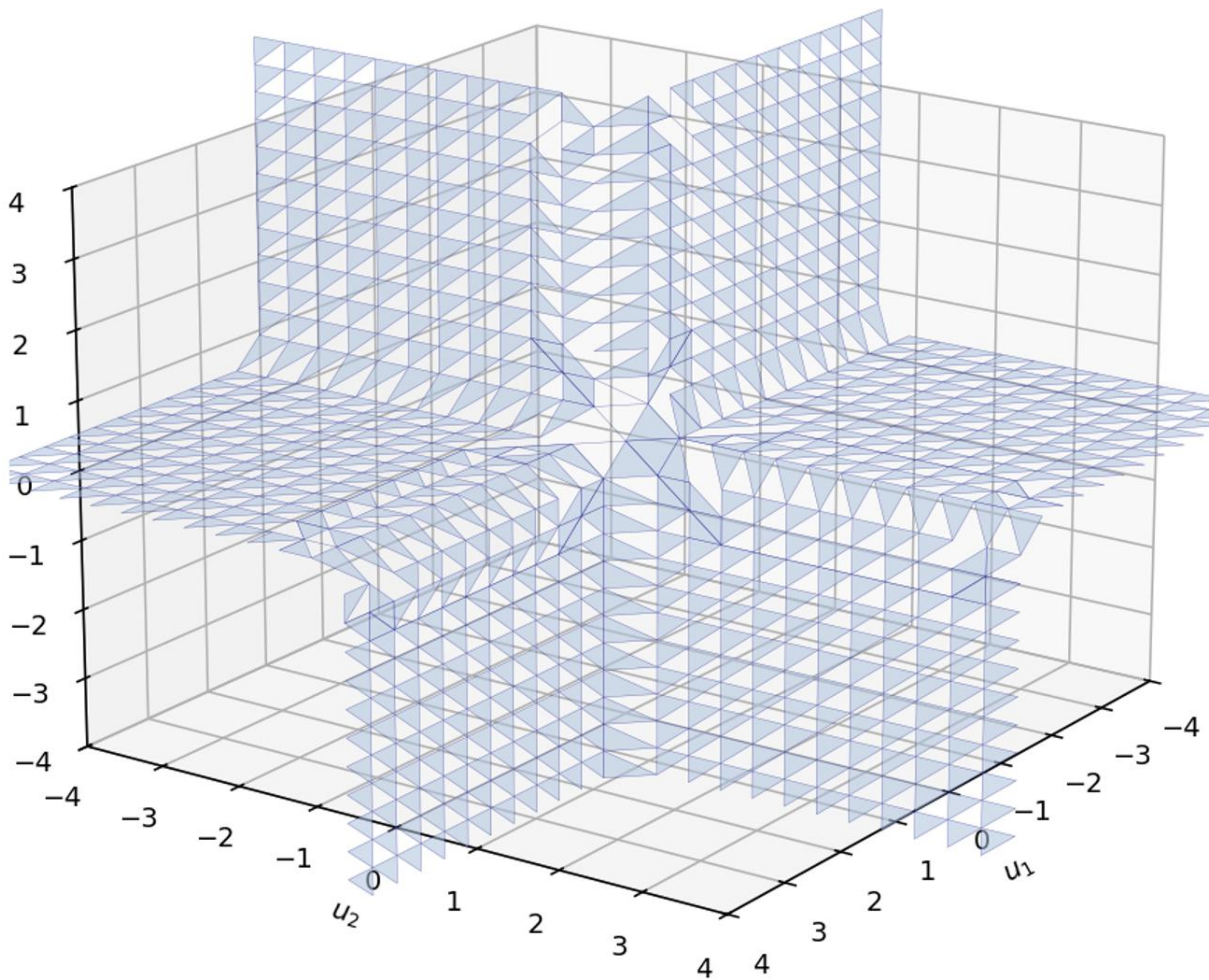
PIPELINE:

- marching cubes ONCE on a certified kernel fixed point (topology pinned)
- $\rightarrow$ JIT'd residual + per-surface FD Jacobian (block-diagonal)
- $\rightarrow$ Levenberg trust-region with per-vertex step caps
- $\rightarrow$ topology guard (no face flips allowed)
- $\rightarrow$ strict- $h=0$ fixed point

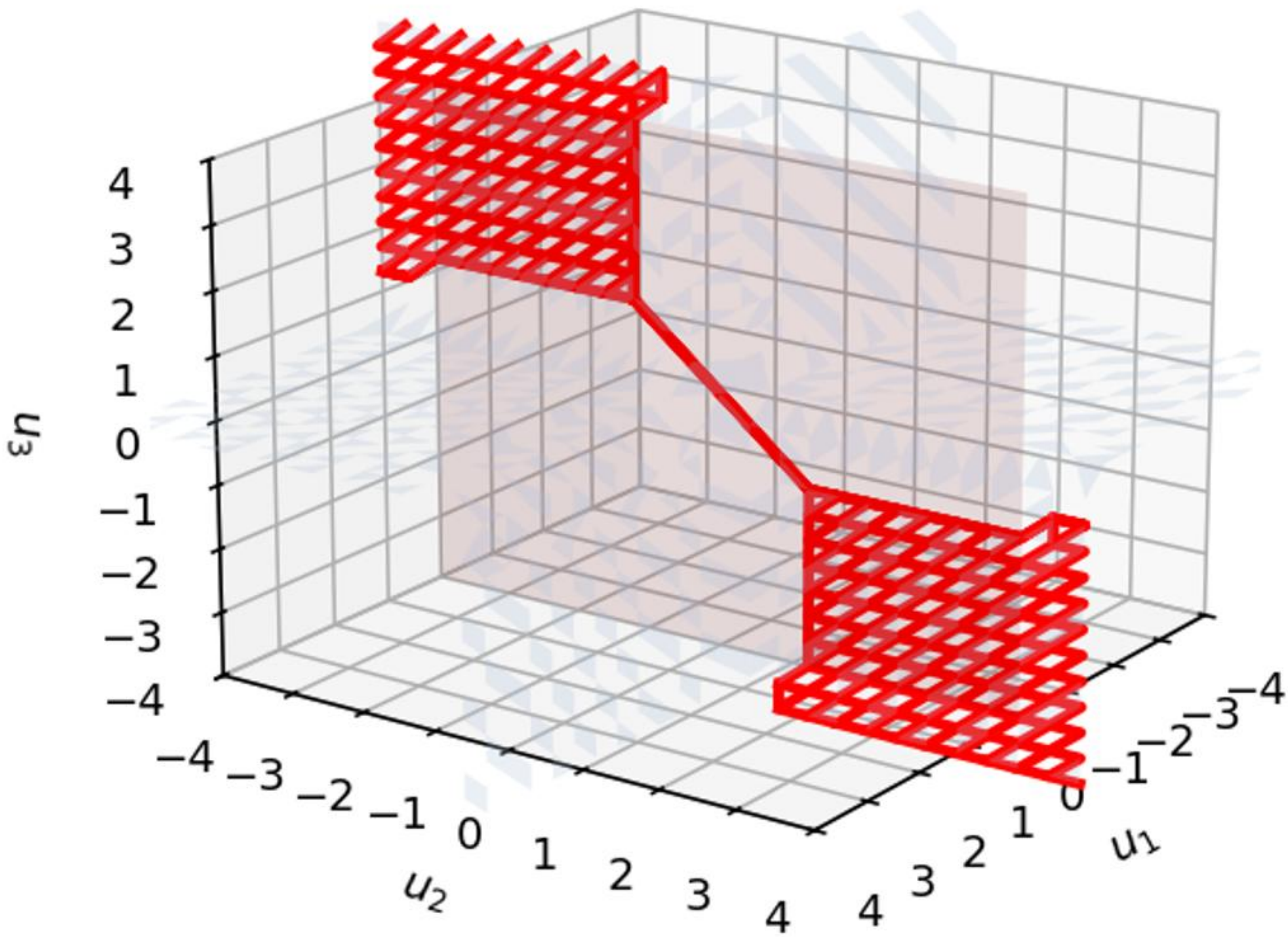
G1 -- Level surfaces $\{P^* = p\}$ for $p \in \{0.25, 0.4, 0.5, 0.6, 0.75\}$
Each surface carries its own endogenous grid (mesh vertices)



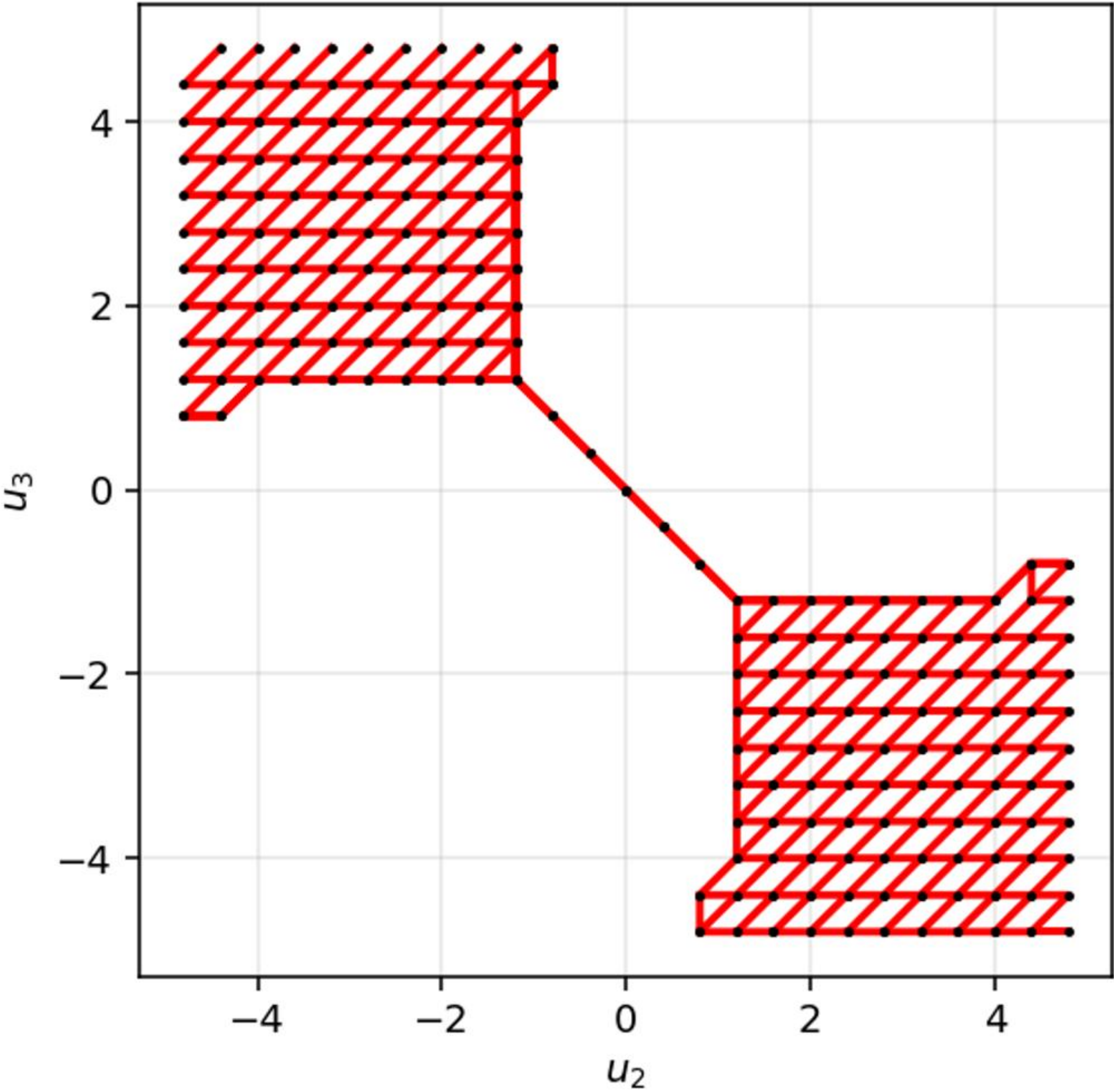
G2 -- The $p = 0.5$ surface as a triangle mesh
957 vertices (UNKNOWN), 1768 faces (topology FIXED)



G3 -- Hyperplane slice $\{u_1 = 0\}$ of the $p = 0.5$ surface
Red curve = the strict- $h = 0$ integration domain

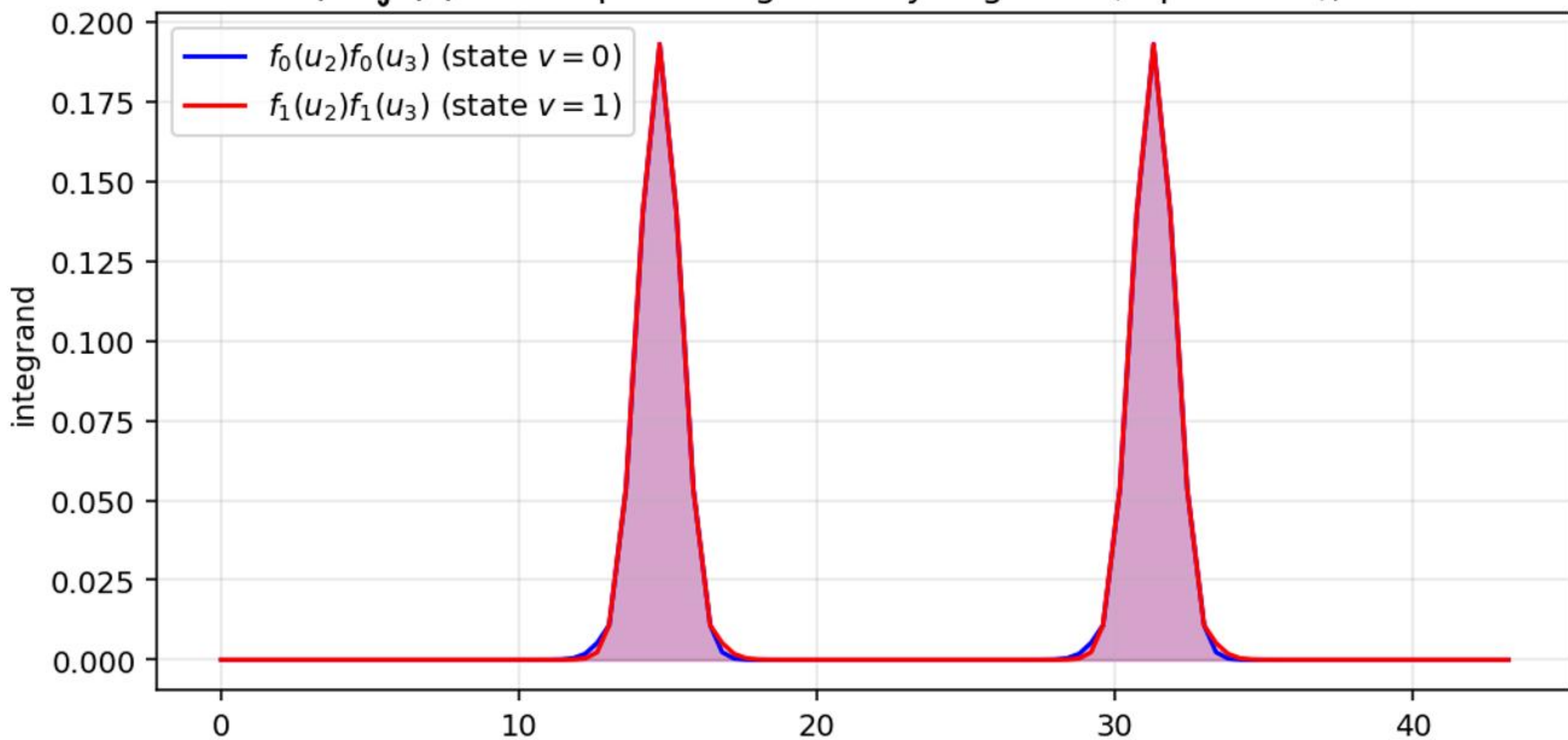


Same curve in the (u_2, u_3) plane -- a piecewise-linear polyline
(black dots = segment endpoints from face crossings)

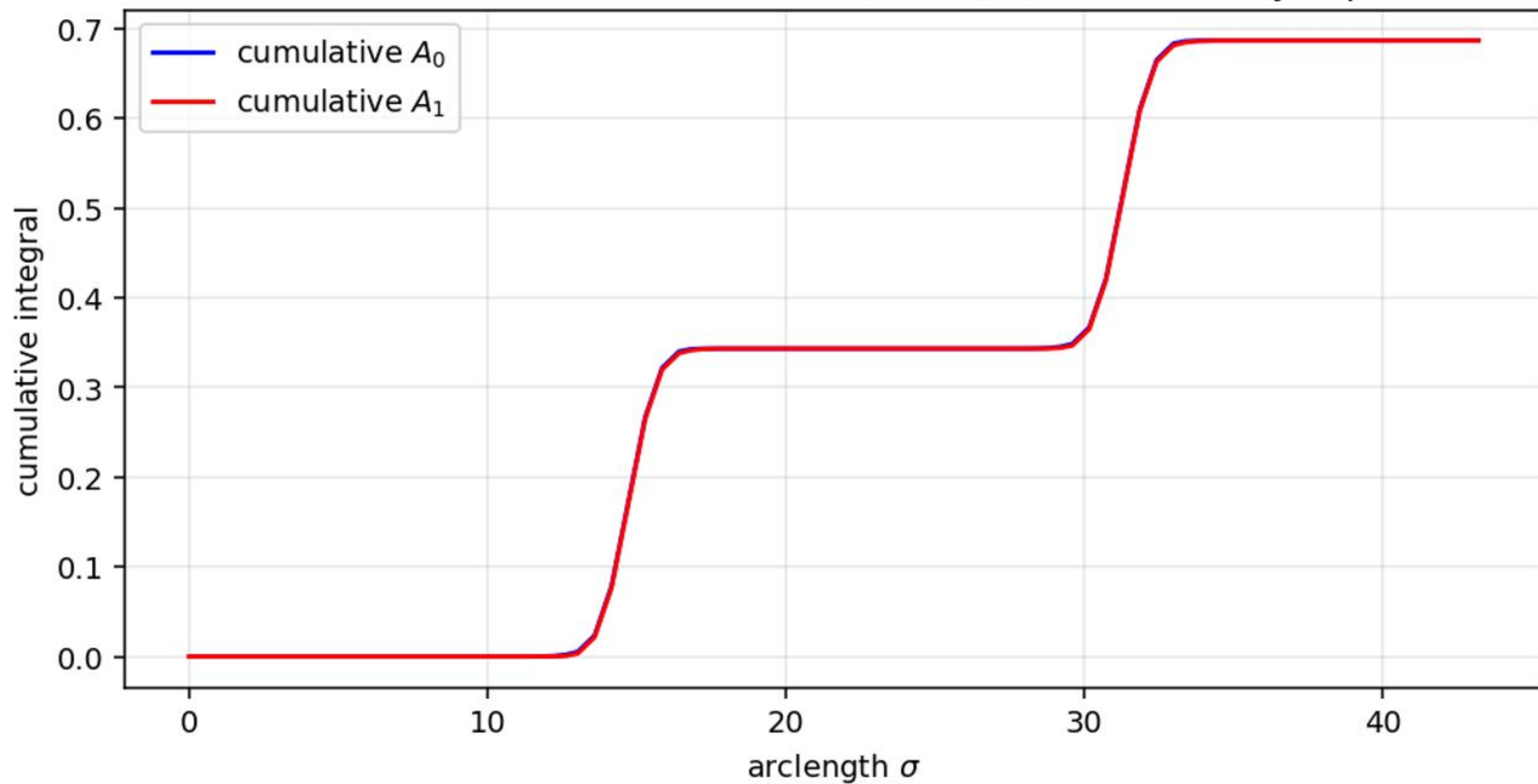


G4 -- Evidence integrands along the curve

$A_v = \int f_v f_v d\sigma$ computed segment-by-segment (2-pt Gauss), $h \equiv 0$



Cumulative evidence -- the likelihood ratio A_1/A_0 feeds the Bayes posterior



G5 -- The per-vertex residual

For each vertex $v = (v_1, v_2, v_3)$ of surface S_m (price level p_m):

1. Three slices through the SAME surface:

agent 1: curve $= S_m \cap \{u_1 = v_1\} \rightarrow (A_0^{(1)}, A_1^{(1)})$

agent 2: curve $= S_m \cap \{u_2 = v_2\} \rightarrow (A_0^{(2)}, A_1^{(2)})$

agent 3: curve $= S_m \cap \{u_3 = v_3\} \rightarrow (A_0^{(3)}, A_1^{(3)})$

2. Bayes posterior per agent:

$$\mu_k = \frac{f_1(v_k) A_1^{(k)}}{f_0(v_k) A_0^{(k)} + f_1(v_k) A_1^{(k)}}$$

3. CRRA market clearing (bisection, 60 iters):

$$p^{clear}(v) = \text{unique } p: \sum_k x_k^{CRRA}(\mu_k, p; \gamma) = 0$$

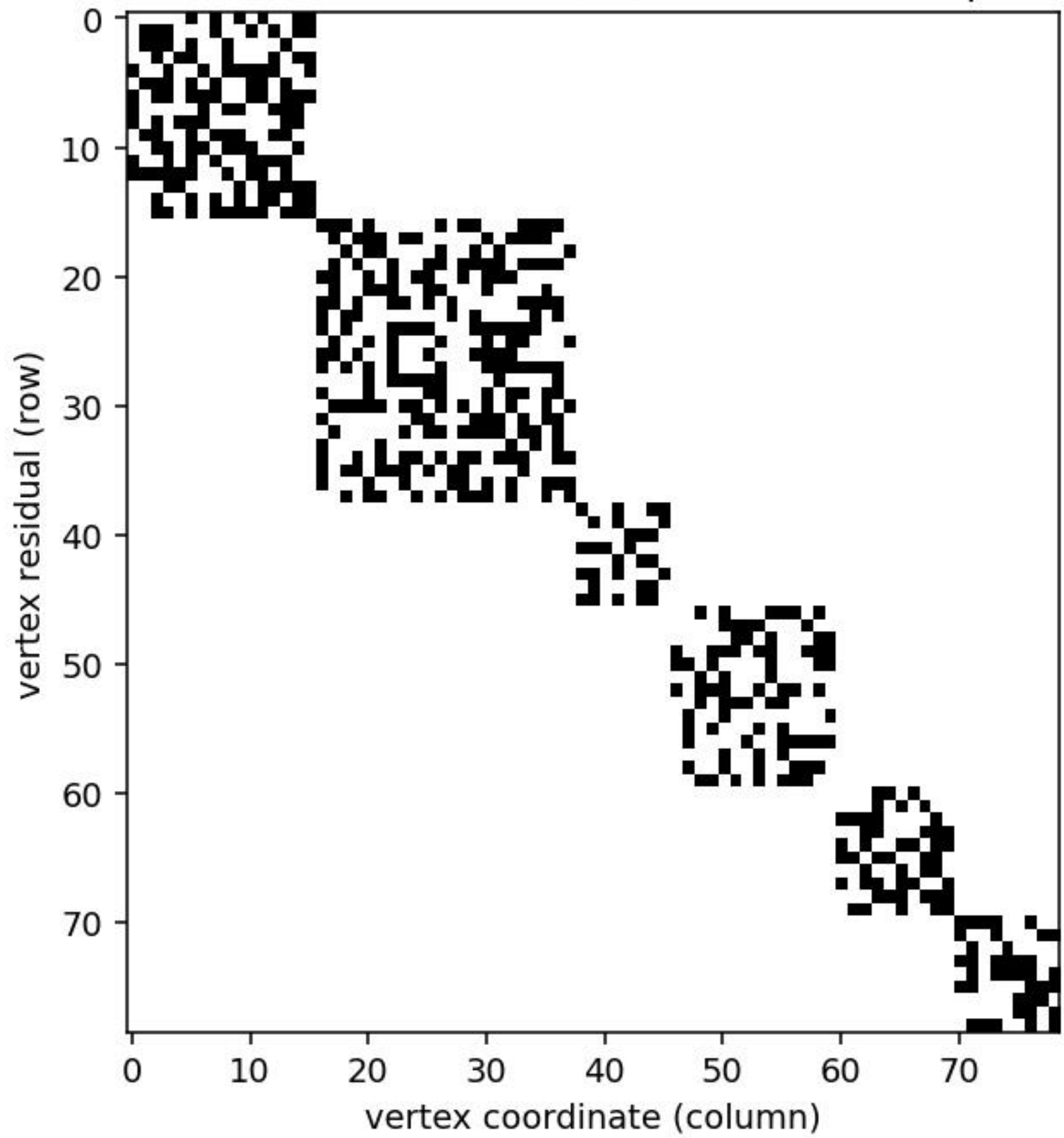
4. Residual:

$$r(v) = p^{clear}(v) - p_m$$

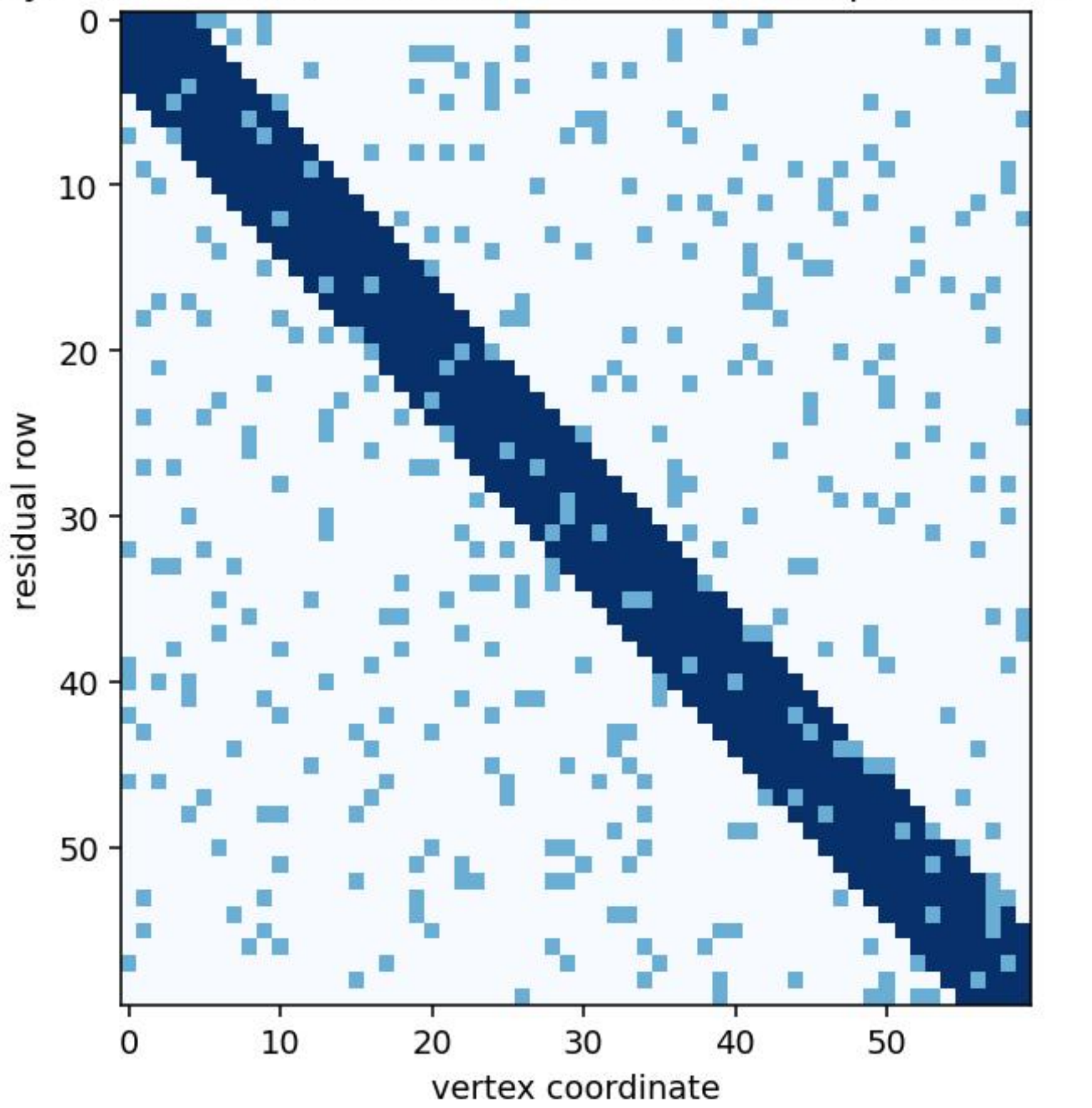
At a strict- $h = 0$ equilibrium, every vertex of every surface clears exactly at its own price level: $r \equiv 0$.

The unknowns are the vertex coordinates themselves: when LM moves a vertex, it changes (a) its own slice planes, (b) the curve geometry of every slice that crosses its incident faces -- both effects are smooth in the coordinates because the mesh topology never changes.

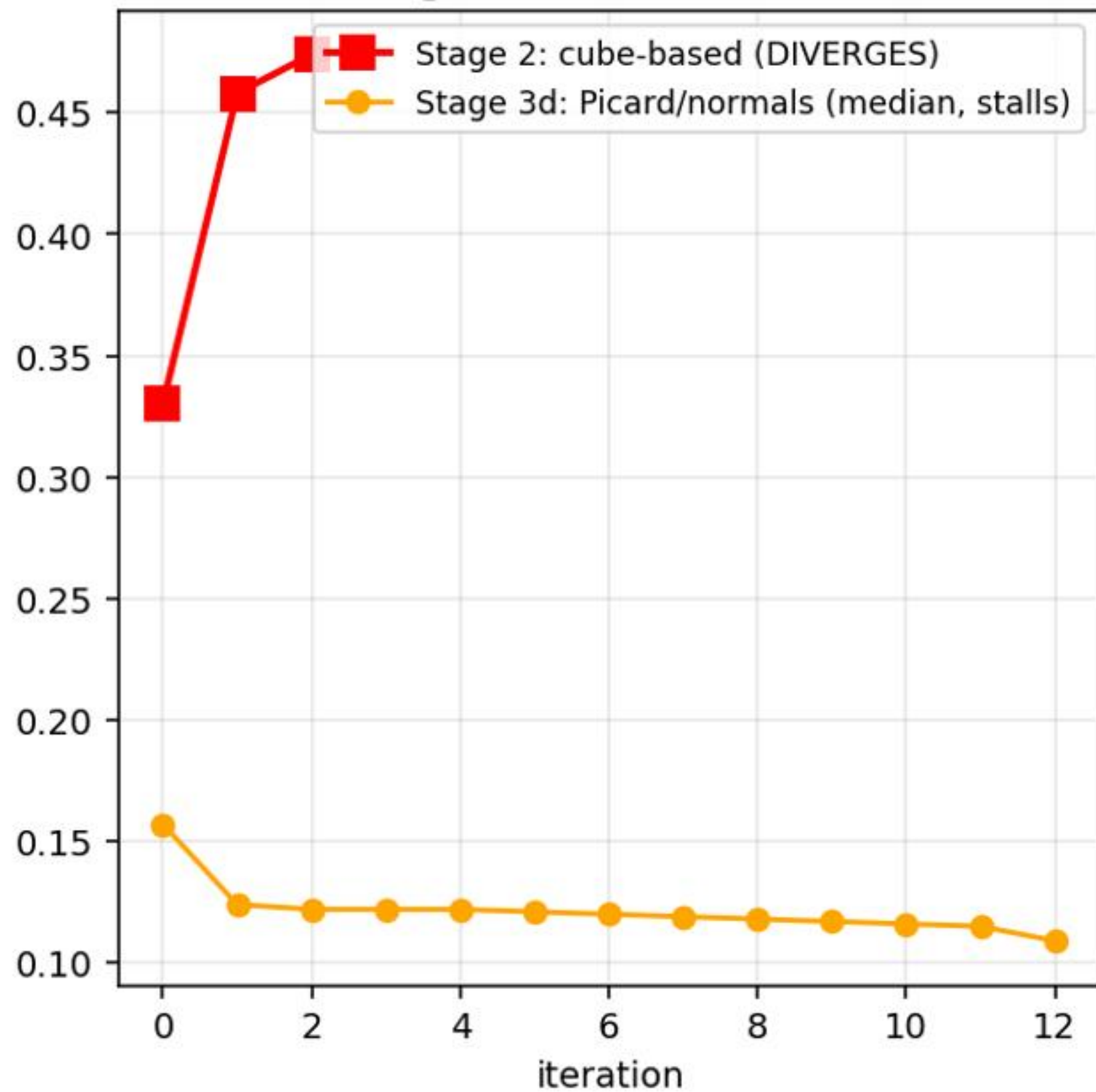
G6a -- Global Jacobian is block-diagonal by surface
(surfaces share no residuals; blocks solved independently)



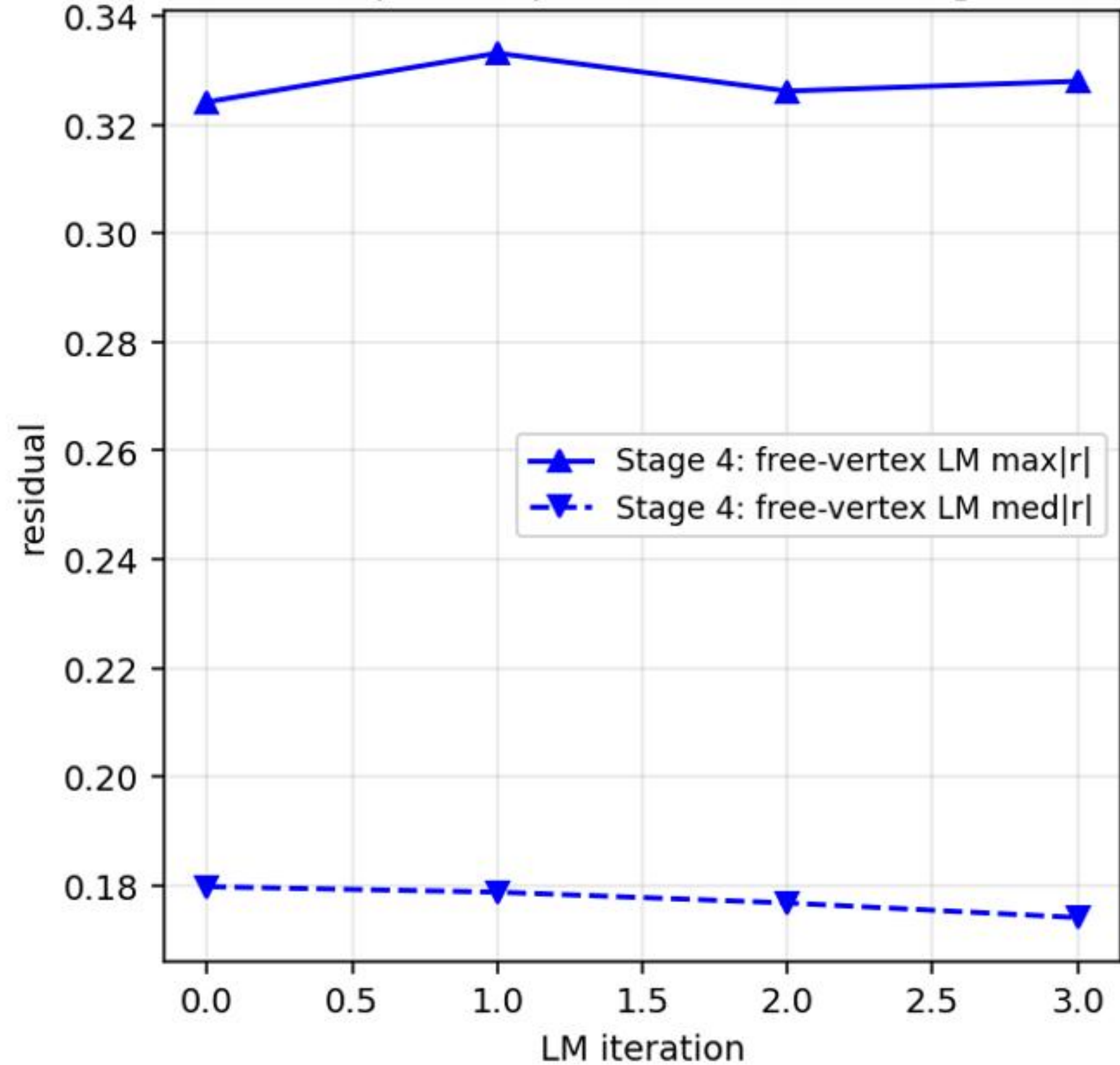
G6b -- WITHIN a block (Stage 5 sparse-FD):
column w affects only rows whose slice planes cross
faces incident to vertex w -- $O(1)$ rows per column



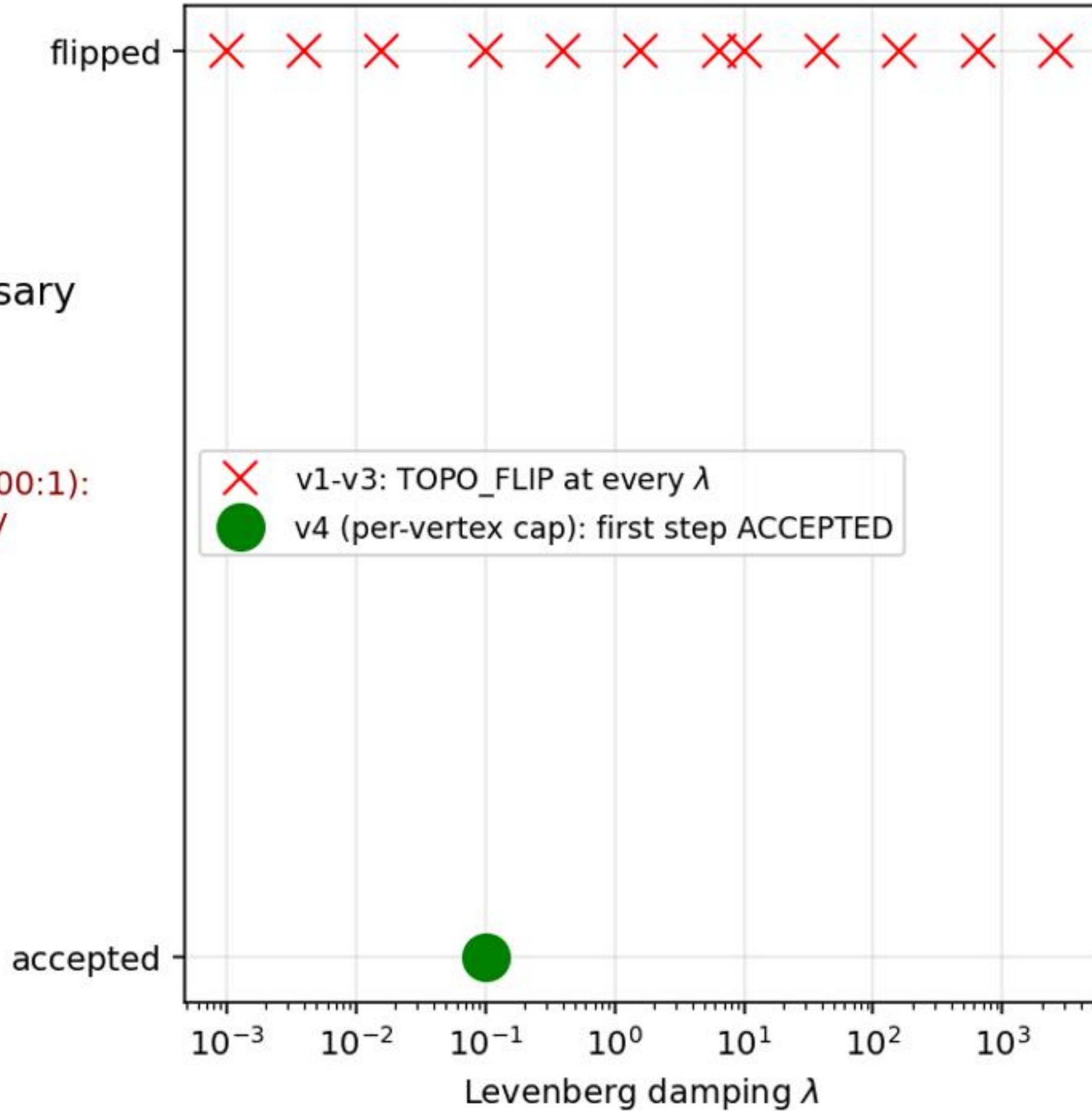
G7a -- Why representation matters:
cube diverges; constrained mesh stalls



G7b -- Stage 4 free-vertex LM (running):
steps accepted, cost decreasing



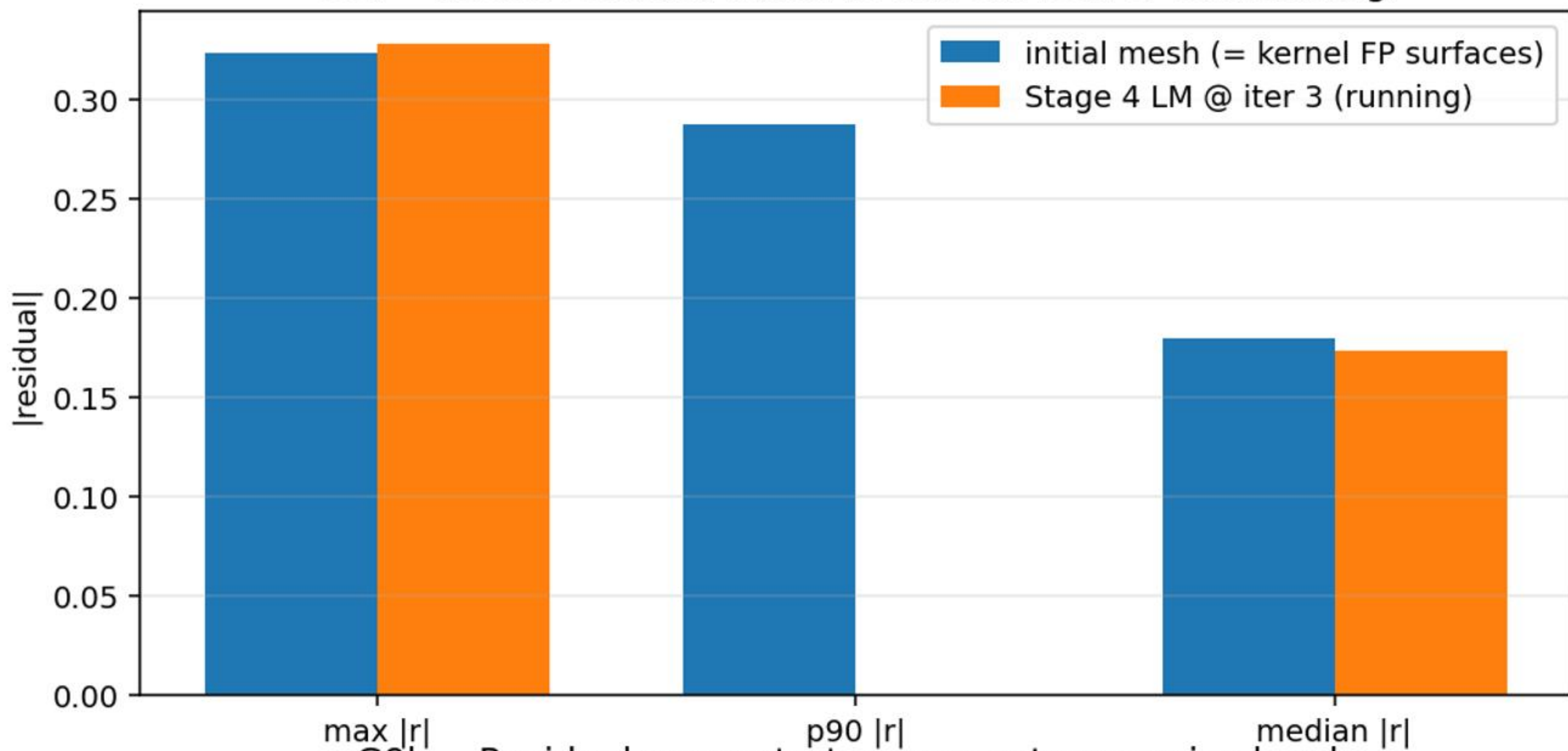
G8b -- Damping alone cannot prevent flips; per-vertex caps can



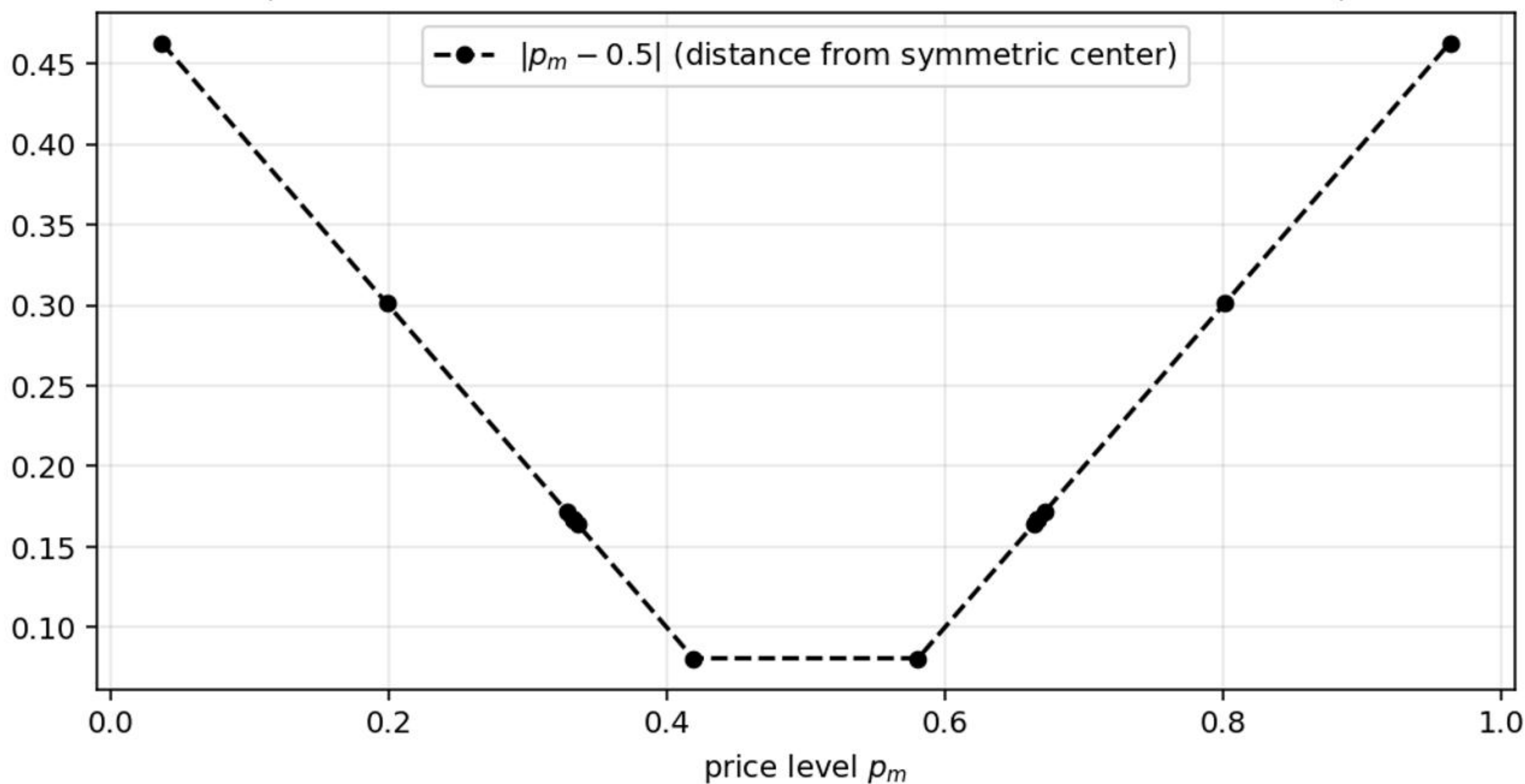
G8a -- Why per-vertex local step caps were necessary



G9a -- Vertex residual: initial vs current (solver still running)



G9b -- Residual concentrates near extreme price levels
(tail surfaces are smallest + most curved -- hardest to clear)



Status and roadmap

WHAT IS PROVEN (committed in repo):

- Stage 1c: the strict- $h=0$ forward operator is correct
 - matches kernel operator to $O(h^2)$ in bulk, $O(h)$ tails
 - density unit test (V1, earlier CDF work): $O(du^2)$ convergence
- Stage 2: the CUBE cannot host this operator (re-triangulation jumps); diverges 0.33 $\rightarrow$ 0.47. Negative result, documented.
- Stage 3: moving-mesh representation is correct; constrained (normal-only) motion is insufficient. JIT residual: 0.6 s/eval (was 90 min in Python).
- Stage 4: free-vertex LM with per-vertex caps + topology guard ACCEPTS steps and decreases cost (running now). Dense block FD Jacobian = 8 min/iter $\rightarrow$ too slow to converge in-session, but the dynamics is finally right.
- Stage 5: sparse-FD Jacobian (locality: a vertex's motion affects only slices crossing its ~ 6 incident faces) -- targeted < 60 s/iter. Being built by a parallel agent.

WHAT REMAINS FOR A PUBLISHABLE STRICT- $h=0$ LADDER:

1. Stage 5 convergence at the anchor cell ($\tau=2$, $\gamma=0.098$)
2. Deficit of the converged mesh vs deep-ladder continuum estimate $d_{\inf} = 0.268 \pm 0.010$ (the validation gate)
3. Mesh refinement study (M price levels, mesh resolution)
4. Parameter ladder with mesh warm-starts + topology re-extraction when the guard fires

The method is sound; the engineering risk is concentrated in mesh quality management (slivers from marching cubes should be improved by isotropic remeshing at initialization).